USHPI Granger Causality Analysis Case Study Hook Document

Imagine you work as a data analyst for a housing policy research organization. Your team has been asked a fairly straightforward question: can we use wage data to get ahead of fluctuations in home prices? It sounds simple, but answering that question may take a bit of work.

You have access to two time series which are from the Federal Reserve Bank of St. Louis. They are monthly average hourly wage data and the U.S. National Home Price Index (USNHPI). Your data covers issues like the 2008 financial crisis and the COVID-19 pandemic. There is a lot going on in this dataset. Part of your job is to make sense of it.

Your method will be Granger causality. The idea is that if knowing the past values of one variable helps you better predict another variable, then the first variable will Granger-cause the second. This is not the same as saying wages actually cause home prices to rise. Your analysis should make that distinction clear.

First, you will need to clean and merge the two datasets. Then, check that the data behaves the way the method requires. And finally, run the actual Granger causality test. Once you have your results, you will have to explain what they mean and what they do not mean, in a way that someone without a statistics background could follow.

At the end of this case study, you will put together a presentation and a GitHub repository that walks through your entire process, from pulling the raw data to drawing conclusions. The rubric has all the specifics on what your deliverables should look like.

Clone the repository and start by reading the README. From there, review the supplemental readings on Granger causality before writing any code. The data is freely available from FRED and the repo includes instructions for how to download it.

Housing affordability is one of the most talked-about issues in the country right now. The data is there. See what it actually tells you.

Everything you will need can be found here:
https://github.com/jonahsjlee/DS-4002-CS3.git